Project 3: Monocular Metric Depth Estimation & Video Consistency

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Spring 2027



1 Overview

Core Task: Learn stereo geometry on Middlebury, evaluate one pretrained metric-depth model on NYU Depth V2, and compare image-based and video-based depth on a provided example clip. Then implement ONE improvement and demonstrate a simple application. Static images provide quantitative evaluation; the video provides qualitative comparison.

Key Concepts:

- **Projective Geometry:** Disparity, focal length, camera intrinsics, and metric scale.
- **Monocular Depth:** Relative depth versus zero-shot metric depth.
- **Foundation Models:** Large-scale pretraining and cross-dataset generalization.
- **Video Consistency:** Temporal propagation, optical flow, scale drift, and occlusion.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report (Mandatory):**
  - **Format:** Use the CVPR LaTeX Template. The length must be 6–8 pages (excluding references).CVPR template

- **ArXiv Upload (Optional):** You can upload your final report to arXiv. Remember to change the "REVIEW version" to "CAMERA-READY version" in the template. Please clearly indicate your arXiv ID on the first page of the Canvas submission.
- **Content Requirements:**

1. **Abstract & Introduction:** Background, motivation, problem definition, and solution overview. Explicitly include your public GitHub repository link at the end of the abstract.
2. **Related Work:** Review and cite at least all papers listed in the Recommended Reading List (Section 4). Expanding beyond this list is encouraged.
3. **Method:** Explain the technical roadmap in your own words. Use high-quality pipeline diagrams and visual examples to support the explanation.
4. **Experiments:** Present quantitative results in tables, qualitative comparisons in figures, ablation studies, and categorized failure cases.
5. **Conclusion:** Summarize the findings, limitations, and specific directions for future work.

- **Code (Mandatory):** Upload the code to a public GitHub repository with a clear README.md. Include dependencies, data preparation, model weights, training and evaluation commands, and representative visual results. Do not submit raw code files to Canvas.
- **Depth Demo & Video (Mandatory):** Submit a side-by-side RGB/depth comparison on the first 100 frames of davis_rollercoaster.mp4 from the linked repository in Section 6.1. Include one simple application output, such as depth-aware background blur or a point-cloud view, from a provided image or clip. A recorded demo is sufficient. Pack outputs into depth_results.zip.

Important Tips for Success

- **Method Flexibility:** You may follow the suggested roadmap or design another technically sound method. Strict adherence to the Implementation Roadmap is not required if the core task and evaluation protocol are fulfilled.
- **Completeness:** Complete Parts 1 and 2, then choose ONE improvement in Part 3. Compare it with the unchanged method. If accuracy does not improve, explain why using controlled experiments and failure cases. Ignoring Part 3 will result in an "Incomplete" grade.
- **Visuals:** Clean pipeline diagrams, prediction overlays, zoomed comparisons, and failure visualizations are essential in a Computer Vision course and will substantially affect the presentation score.
- **Dataset Scope:** Use Middlebury for the stereo exercise, NYU Depth V2 for quantitative monocular evaluation, and the specified repository video for demonstration and qualitative comparison only. No self-recorded clips or additional datasets are required. Assessment emphasizes the idea, measured effect, and analysis.
- **Fair Comparison:** Use the same test images, evaluation masks, output size, and metrics for the original and improved method. Preserve each pretrained model's required preprocessing and report its input size, runtime, and peak GPU memory.
- **Citations:** Cite all datasets, pretrained models, external repositories, and borrowed implementation components correctly.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1:** Literature Review. Start with stereo geometry, Depth Anything V2 [5], your chosen metric-depth model [7 or 8], and Video Depth Anything [10]. Optical-flow methods are relevant only if you choose a temporal-fusion improvement.
2. **Step 2: Code Familiarization.** Clone and run the official repositories for Depth Anything V2 and Video Depth Anything. Run their provided demos first, then verify the input/output format and pretrained checkpoint on one sample.

3. **Step 3:** Experimentation. Follow the download links in Section 6.1. First run one image, then evaluate the static test set. Finally process the specified 100-frame clip. Save floating-point depth predictions as well as colored visualizations.
4. **Step 4: Writing & Submission.** Organize quantitative results, ablations, and failure cases into tables and figures. Write the report with the CVPR template, cite all references, and submit the required files to Canvas.

4 Recommended Reading List

- **Classical Geometry:** Semi-Global Matching [1], KITTI benchmark [2]
- **Monocular Foundations:** DPT [3], Depth Anything [4], Depth Anything V2 [5]
- **Metric Depth:** UniDepth [6], Depth Pro [7], UniDepthV2 [8]
- **Video Depth & Motion:** RAFT [9], Video Depth Anything [10]

5 Implementation Roadmap

5.1 Part 1: The Baseline – "Geometry & Frame-Wise Depth"

Goal: Understand the geometric meaning of depth and establish frame-wise baselines before using temporal models.

1. Classical Stereo: Disparity to Depth

- **Step 1:** Rectified Stereo. Run OpenCV StereoBM or StereoSGBM on the 15 Middlebury trainQ pairs. Display disparity and invalid regions; left-right consistency is an optional diagnostic.
- **Step 2:** Metric Conversion. Use each trainQ/calib.txt: depth in meters equals focal length times baseline divided by (disparity + doffs), then divided by 1000 because baseline is in millimeters. Divide OpenCV's fixed-point disparity by 16 first. Use the quarter-resolution calibration and exclude invalid disparities. | Calibration definition

2. Monocular Frame Baseline

- **Relative Depth:** Run Depth Anything V2-Small independently on the same 100 video frames. This is the frame-wise video baseline; it is not used as a metric-depth prediction on NYU.
- **Scale Convention:** Label this output "relative depth." Do not estimate meters or align it using test ground truth. Metric-depth evaluation in Part 2 uses a separate metric model.
- **Temporal Diagnosis:** Compare the frame-wise results with Video Depth Anything on the provided clip. Describe visible flicker, unstable edges, and motion artifacts. No camera-pose annotations or temporal ground truth are needed.

Expected Result: Frame-wise models should recover convincing scene structure but may have incorrect metric scale, unstable boundaries, and visible temporal flicker.

5.2 Part 2: SOTA Reproduction – "Metric & Temporally Consistent Depth"

Goal: Reproduce recent open models for zero-shot metric depth and long-video consistent depth without training a foundation model from scratch.

1. Single-Image Metric Depth: UniDepthV2 [8] or Depth Pro [7]

- **Requirement:** Choose ONE metric model; UniDepthV2 ViT-S is the recommended starting point. Load UniDepthV2.from_pretrained("lpiccinelli/unidepth-v2-vits14") and follow the repository's infer(rgb) example. Depth Pro is an alternative. Use RGB-only inference and evaluate raw metric output on NYU without test-set scale fitting. | UniDepthV2 code | ViT-S checkpoint | Alternative Depth Pro
- **Key Learning Point:** Compare the chosen metric model with your improvement using the same NYU protocol. Explain errors in absolute scale and object boundaries. Training a foundation model is not required.

2. Video Depth: Video Depth Anything [10]

- **System:** Use the Video Depth Anything Small relative-depth checkpoint on the first 100 frames of davis_rollercoaster.mp4. Use offline inference; streaming and metric-video modes are optional.
- **Required Comparison:** Show frame-wise Depth Anything V2-Small and Video Depth Anything-Small on exactly the same frames. Use the same color palette and a fixed normalization for each method over the entire clip. Label outputs as relative depth; colors do not represent a shared metric scale.
- **Resource Rule:** Use Small models and the video repository's default FP16. Start with --encoder vits --max_len 100 --max_res 640 --save_npz. Reduce --input_size if memory is insufficient and report the setting. Keep resolution fixed between a method and its own improvement.

Expected Result: Observe whether video modeling reduces flicker and identify any remaining artifacts. These clips have no provided depth ground truth, so visual observations do not establish metric accuracy.

5.3 Part 3: Exploration – "Calibration, Consistency & Application"

Goal: Choose ONE lightweight improvement. The directions below are alternatives. A static-image refinement with NYU evaluation is the simplest route. A video improvement may instead use controlled qualitative comparisons and runtime measurements; no annotated video collection is required.

- **Direction A (Flow-Guided Fusion):** Warp the previous depth map with RAFT or Farnebäck flow, reject occlusions with forward-backward consistency, and fuse it with the current prediction using confidence weights.
- **Direction B (Metric Scale Calibration):** Fit one global scale using only the NYU training/development images, then freeze it for the test split. Explain whether the improvement generalizes; do not fit a separate scale to test ground truth.
- **Direction C (Edge-Aware Refinement):** Use RGB gradients, semantic boundaries, or segmentation masks to sharpen depth discontinuities without copying texture into depth.
- **Direction D (Uncertainty-Aware Smoothing):** Estimate uncertainty from model confidence, test-time augmentations, or cross-model disagreement. Smooth only uncertain and temporally inconsistent regions.
- **Direction E (Efficient Deployment):** Export a Small model to ONNX/Core ML, apply quantization, or use dynamic resolution. Report speed, memory, and accuracy trade-offs.
- **Direction F (Depth-Aware Application):** Improve a simple effect such as background blur or depth-based object ordering. Compare it with the same application using the unchanged depth model. Complex interfaces, extra sensors, and new annotated data are optional.

6 Dataset & Evaluation

6.1 Datasets and Required Downloads

1. **Middlebury Stereo v3 (Stereo Exercise):** Download the quarter-resolution input ZIP (~30 MB) and left-view ground-truth ZIP (~13 MB), without a form. Use only the 15 trainQ pairs, their calib.txt files, and disp0GT.pfm ground truth; the benchmark test set is not required. | Input ZIP | Ground-truth ZIP | Benchmark instructions
2. **NYU Depth V2 (Metric Evaluation):** Download the labeled .mat file (~2.8 GB) and splits.mat below, without a form. Use trainNdxs (795 images) for development and testNdxs (654 images) for evaluation; subtract 1 from MATLAB indices in Python. With h5py, read RGB as f["images"][i].transpose(2,1,0) and filled depth in meters as f["depths"][i].T. Keep depth as floating point. The linked loader illustrates .mat access, but use depths and the course protocol below, not its rawDepths/alignment evaluation. No raw video collection is needed. | Labeled data | Official split file | Dataset fields | Python loader example
3. **Video and Small Checkpoints:** The required clip is assets/example_videos/davis_rollercoaster.mp4, included when cloning Video Depth Anything. Use its first 100 frames at the original frame rate. Download the two relative-depth Small weights below into each repository's checkpoints/ folder. Follow the linked README installation and inference commands; for video use the flags in Part 2. Other example clips are optional. | Example videos | Clip download | Frame-wise code | Frame-wise Small weights | Video code | Video Small weights

6.2 Metrics (Mandatory)

- **Metric Depth Accuracy:** On NYU, report AbsRel (main metric), RMSE, and δ1. Resize predictions to 480 × 640. Evaluate the Python crop [45:471, 41:601] where ground-truth depth is between 0.001 and 10 meters; clip predictions to this range. Average per-image metrics equally over all 654 test images. Use no scale alignment. Compare your own runs under this course protocol rather than claiming an exact reproduction of a paper's score.
- **Stereo Exercise:** Report mean absolute disparity error in pixels and the fraction of valid predictions on finite, positive trainQ ground-truth disparities. State the invalid-prediction rule. This exercise is separate from the NYU metric-model comparison.
- **Video Comparison:** Qualitative only. Show the same frames and discuss temporal smoothness, edge stability, and motion artifacts. Numerical temporal error, scale drift, optical-flow evaluation, and depth-edge precision/recall are optional.
- **Efficiency:** Report inference time per image/frame and peak GPU memory, with hardware, model variant, precision, and input size.
- **Qualitative Evaluation:** Show RGB, predicted depth, the chosen application output, and three representative difficult cases actually present in the supplied data. Explain limitations rather than requiring particular scene types.

Note: Never normalize each video frame independently; this can conceal flicker. Freeze normalization over the whole clip, and use the unchanged method's normalization for its improvement.

References

[1] Hirschmüller, H. "Stereo Processing by Semiglobal Matching and Mutual Information." TPAMI 2008.
[2] Geiger, A., et al. "Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite." CVPR 2012.
[3] Ranftl, R., et al. "Vision Transformers for Dense Prediction." ICCV 2021.
[4] Yang, L., et al. "Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data." CVPR 2024.
[5] Yang, L., et al. "Depth Anything V2." NeurIPS 2024.
[6] Piccinelli, L., et al. "UniDepth: Universal Monocular Metric Depth Estimation." CVPR 2024.
[7] Bochkovskii, A., et al. "Depth Pro: Sharp Monocular Metric Depth in Less Than a Second." ICLR 2025.
[8] Piccinelli, L., et al. "UniDepthV2: Universal Monocular Metric Depth Estimation Made Simpler." arXiv:2502.20110, 2025.
[9] Teed, Z., and Deng, J. "RAFT: Recurrent All-Pairs Field Transforms for Optical Flow." ECCV 2020.
[10] Chen, S., et al. "Video Depth Anything: Consistent Depth Estimation for Super-Long Videos." CVPR 2025.
[11] Sturm, J., et al. "A Benchmark for the Evaluation of RGB-D SLAM Systems." IROS 2012.
[12] Silberman, N., et al. "Indoor Segmentation and Support Inference from RGBD Images." ECCV 2012.